

Ollama, a framework for building and running LLMs

Ollama is a specialised platform that facilitates the deployment and management of large language models (LLMs) in local environments. This tool is particularly valuable for developers and researchers who need to fine-tune and run models without relying on cloud-based infrastructure, ensuring both privacy and control over computational resources.

The installation and working process typically involves downloading the Ollama package and setting up the necessary dependencies.

Once Ollama is downloaded and installed, deploying an LLM locally is straightforward. The platform provides a set of commands that allow users to load, fine-tune, and interact with language models.

To initiate the deployment of a model, you can use the following command:

```
CommandPrompt > ollama run llama2
```

In this command, `llama2` is a placeholder for the specific model you wish to deploy. Ollama supports

a range of models, and this command will load the selected model into the local environment. The deployment process typically involves initialising the model with the necessary computational resources, such as GPU or CPU, depending on the specifications of your machine.

After deploying the model, Ollama provides an interactive interface for running queries and generating text. The platform supports a variety of inputs, enabling users to interact with the model in a dynamic and intuitive manner. For example, you can execute the following command to interact with the model:

```
CommandPrompt > ollama run llama2
-p "Explain the concept of quantum computing."
```

This command prompts the `llama2` model to generate a response based on the provided input. The `-p` flag indicates the prompt that is being passed to the model. Ollama's interface is designed to handle complex queries and can be customised to suit the specific requirements of different applications.

Popular platforms for deploying LLMs

Cloud-based platforms

Amazon Web Services (AWS)
Google Cloud Platform (GCP)
Microsoft Azure

On-premises and hybrid platforms

NVIDIA DGX Systems
Red Hat OpenShift

Edge deployment platforms

NVIDIA Jetson
Google Coral

Open source platforms

TensorFlow Serving
ONNX Runtime

Specialised AI platforms

Hugging Face Inference API
Banana.dev

Ollama also allows users to fine-tune models according to their specific needs. Fine-tuning involves adjusting the parameters of the model to better align with a particular dataset or task. This process can be initiated with the

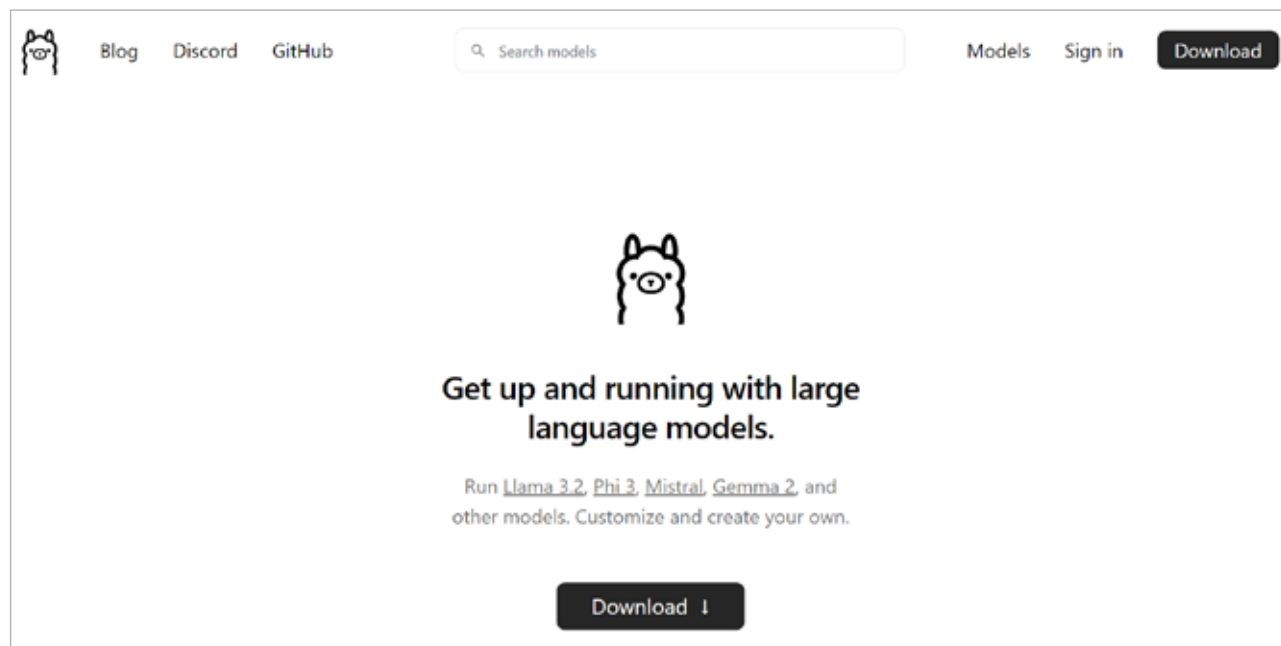


Figure 2: Platform of Ollama for assorted LLM models